

ABSTRACT

Predictive maintenance aims to schedule repairs from the actual condition of a machine rather than a fixed calendar, which requires estimating how many operating cycles a component has left before failure — its Remaining Useful Life (RUL). Most established methods, from linear regression to deep neural networks, report RUL as a single number. For a maintenance decision this is not enough: a prediction of fifty cycles calls for very different action depending on whether it is certain to within three cycles or forty. It is the uncertainty, not the point value, that a safe decision depends on, and it is precisely what a point estimate discards.

This thesis develops a probabilistic digital twin for turbofan engine degradation built on a particle filter, a Sequential Monte Carlo method for Bayesian state estimation. A scalar health index is tracked as a hidden state from noisy sensor readings, and its posterior is projected forward to the failure threshold to produce a full RUL *distribution* — a median together with a 90% credible interval — rather than a single value. Every modelling choice, from the health index to the exponential degradation dynamics, is fitted from data and its fit quality reported. The pipeline is evaluated on the NASA C-MAPSS benchmark (sub-datasets FD001 and FD003) over the full 100-engine test fleets, on both point accuracy and, separately, interval calibration.

The results are reported honestly in both directions. A simple regression baseline remains the sharper point predictor, but it offers no measure of confidence at all. The pipeline moves from an earlier configuration whose 90% credible interval covered essentially no test engines (0.000 coverage) to one that covers 0.75 on FD001 and 0.99 on FD003. The largest single component of this improvement is the calibration of the failure threshold from training-fleet data (Section 3.3.2); estimating the degradation parameters from data — by Particle Marginal Metropolis–Hastings and then by a per-engine similarity library — adds a further improvement on both point accuracy (roughly halving the final-cycle error) and calibration. The reported credible intervals therefore empirically cover the truth close to the nominal rate, with a residual imperfection in each direction: mild overconfidence on FD001 and over-conservatism on FD003 under its harder second fault mode.